



深圳市中深光电股份有限公司

SHENZHEN ZHONGSHEN OPTOELECTRONICS STOCK CO.LTD

PRODUCT SPECIFICATION

MODEL: ZS101NI4042J4H8II-B

< ◇ > PRELIMINARY SPECIFICATION

< ◆ > APPROVAL SPECIFICATION

CUSTOMER
APPROVED BY
DATE:

DESIGNED	CHECKED		APPROVED

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REVISION RECORD

<u>REV NO</u>	<u>REV DATE</u>	<u>PAGE</u>	<u>CONTENTS</u>	<u>ISSUER</u>
1.0	2019-9-20	24	First Release	Zhangxianglin
1.1	2019-12-18	4	Modify voltage and current	Zhangxianglin
1.2	2020-5-28	7	Modify Timing Characteristics	Zhangxianglin



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1.0 GENERAL SPECIFICATIONS

ZS101NI4042J4H8II-B is a color active matrix LCD module incorporating amorphous silicon **TFT** (Thin Film Transistor). It is composed of a color TFT-LCD panel, driver IC, FPC and a backlight unit. The module display area contains **800x 1280** pixels. This product accords with RoHS environmental criterion.

Item	Contents	Unit
Screen Diagonal	10.1"	Inch
Viewing direction	Full View	
Shape size	143*228.6*2.5	mm
Display Area	135.36*216.58	mm
Number of Dots	800(RGB) x1280	/
Display Mode	Normally Black	/
Number of color	16.7M	/
Interface	MIPI	
LCM Luminance	300(typ)	cd/m ²
Response Time (Tr+Tf)	25ms (typ)	/
Contrast Ratio	800	/



ABSOLUTE MAXIMUM RATINGS

The following are maximum values which if exceeded may cause faulty operation or damage to the unit.

Item	Symbol	Min	Max	Unit	Note
Digital Supply Voltage	VCI	-0.5	5.0	V	

3.0 ELECTRICAL CHARACTERISTICS

ITEM	SYMBOL	MIN	TYP	MAX	UNIT	NOTE
Digital Power Supply Voltage For LCD	VCI	3.0	3.3	3.6	V	-
Current for Driver	VCI		120	150	MA	

3.1 BACKLIGHT CHARACTERISTICS

Item	Symbol	Min	Typ	Max	Unit	Condition
Forward voltage	Vf	11.2	12.0	13.2	V	If=180mA
Luminance	Lv	250	300	-	cd/m ²	If=180mA
Number of LED	--	32			Piece	--
Connection mode	P	4chips serial *8			--	--



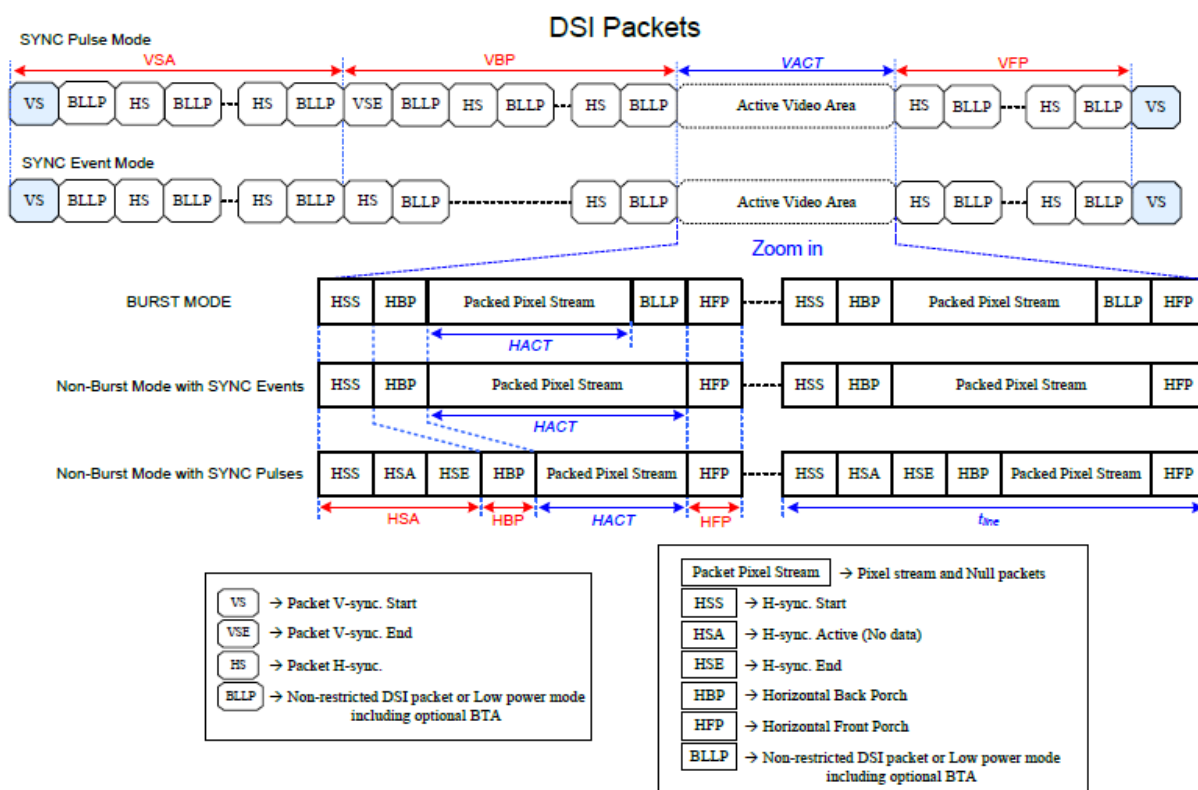
5.0 PINTERFACE PIN CONNECTIONS

Pin No.	Symbol	Description	Remarks
1	NC	Not connection	
2, 3	VCI	Power Supply (3.3V)	
4	GND	Ground	
5	RST	Device reset signal (2.8-3.6V TYP3.3V)	
6	NC	Not connection	
7	GND	Ground	
8	MIPI_ON	MIPI DSI Data Pair 0-Input	
9	MIPI_OP	MIPI DSI Data Pair 0+Input	
10	GND	Ground	
11	MIPI_1N	MIPI DSI Data Pair 1-Input	
12	MIPI_1P	MIPI DSI Data Pair 1+Input	
13	GND	Ground	
14	MIPI_CKN	MIPI DSI Clock- Input	
15	MIPI_CKP	MIPI DSI Clock+ Input	
16	GND	Ground	
17	MIPI_2N	MIPI DSI Data Pair 2-Input	
18	MIPI_2P	MIPI DSI Data Pair 2+Input	
19	GND	Ground	
20	MIPI_3N	MIPI DSI Data Pair 3-Input	
21	MIPI_3P	MIPI DSI Data Pair 3+Input	
22	GND	Ground	
23, 24	NC	Not connection	
25	GND	Ground	
26	NC	Not connection	
27	PWM0 (NC)	Not connection	
28	NC	Not connection	
29	NC	Not connection	
30	GND	Ground	
31, 32	LED-	LED Cathode	
33-38	NC	Not connection	
39, 40	LED+	LED Anode	



6.0 TIMING CHARACTERISTICS OF INPUT SIGNAL

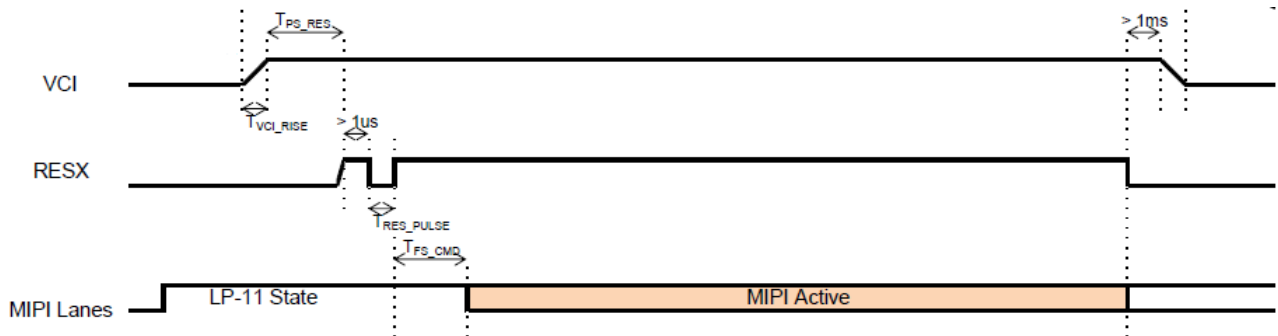
6.1 Timing Characteristics



Parameter	Symbol	Min	Typ	Max	Unit
Horizontal Display	thd	-	800	-	DCLK
HSD Period	th	-	936	-	
HSD Pulse Width	thpw	-	16	-	
HSD Back Porch	thbp	-	60	-	
HSD Front Porch	thfp	-	60	-	
Vertical Display Area	tvd	-	1280	-	DCLK
VSD Period	Tv	-	1304	-	DCLK
VSD Pulse width	tvpw	-	4	-	
VSD Back Porch	tvbp	-	10	-	
VSD Front Porch	tvfp	-	10	-	



6.2 Power on sequence for differential power mode



Symbol	Characteristics	Min.	Typ.	Max.	Units
T_{VCI_RISE}	Case B: VCI Rise time	40	-	-	us
T_{PS_RES}	VDDI/VCI on to Reset high	5	-	-	ms
T_{RES_PULSE}	Reset low pulse time	10	-	-	us
T_{FS_CMD}	Reset to first command	10	-	-	ms

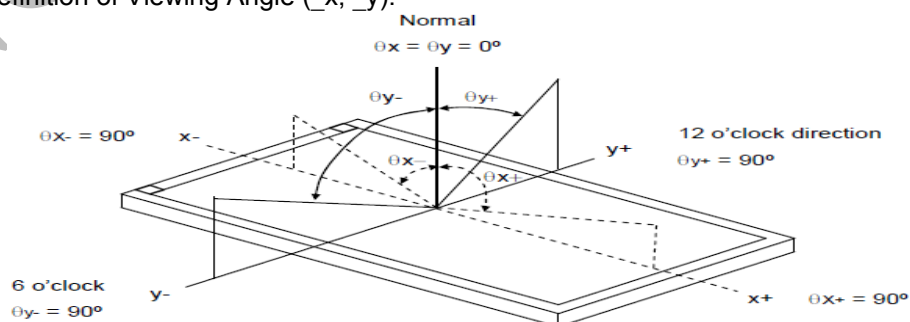


7.0 ELECTRO-OPTICAL CHARACTERISTICS

Ta=25℃

ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNIT	NOTE
Luminance	L	$\theta = 0^\circ$	250	300	--	cd/m ²	Note1
Luminance Uniformity	YU	13 Point	75	80	--	%	Note5
Contrast Ratio	CR	$\theta = 0^\circ$	--	800	--	-	Note3
Response Time	Rr+Tf	$\theta = 0^\circ$	--	25	--	ms	Note4
Viewing Angle K=Contrast Ratio>10	Horizontal	θ_L	CR > 100 $= 0^\circ$	85	--		Note2
		θ_R		85	--		
	Vertical	θ_U		85	--		
		θ_D		85	--		
Color Filter Chromaticity	White	X	$\theta = 0^\circ$	0.255	0.285	0.315	Note1
		Y		0.275	0.305	0.335	
	Red	X	$\theta = 0^\circ$	0.580	0.610	0.640	
		Y		0.320	0.350	0.380	
	Green	X	$\theta = 0^\circ$	0.290	0.320	0.350	
		Y		0.560	0.590	0.620	
	Blue	X	$\theta = 0^\circ$	0.120	0.150	0.180	
		Y		0.020	0.050	0.080	
Color gamut (NTSC ratio)			52	57		%	
colour temperature			6698	7880	9062		

Note (1) Definition of Viewing Angle (θ_x, θ_y):



Note (2) Definition of Contrast Ratio (CR):

The contrast ratio can be calculated by the following expression:



Contrast Ratio (CR) = L255 / L0

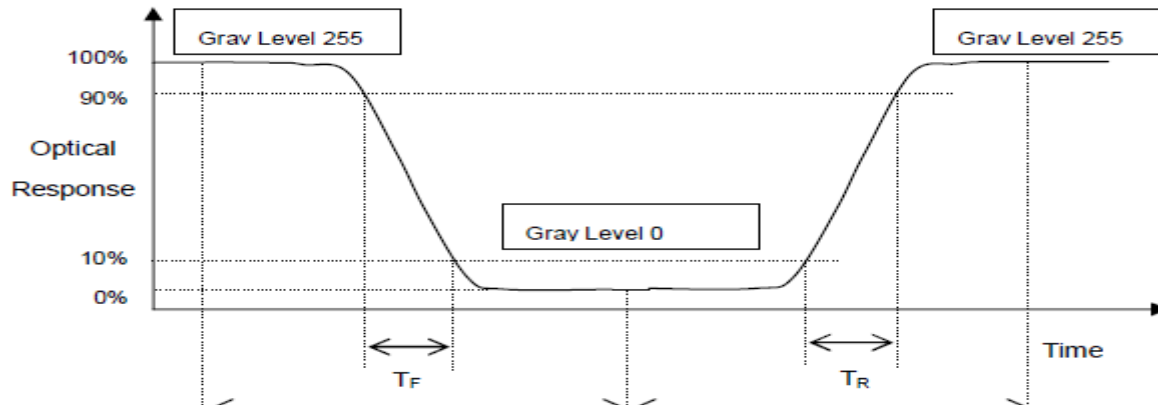
L255: Luminance of gray level 255

L0: Luminance of gray level 0

CR = CR (5)

CR (X) is corresponding to the Contrast Ratio of the point X at Figure in Note (6).

Note (3) Definition of Response Time (T_R , T_F):



Note (4) Definition of Center Point Luminance of White (L_{CP}):

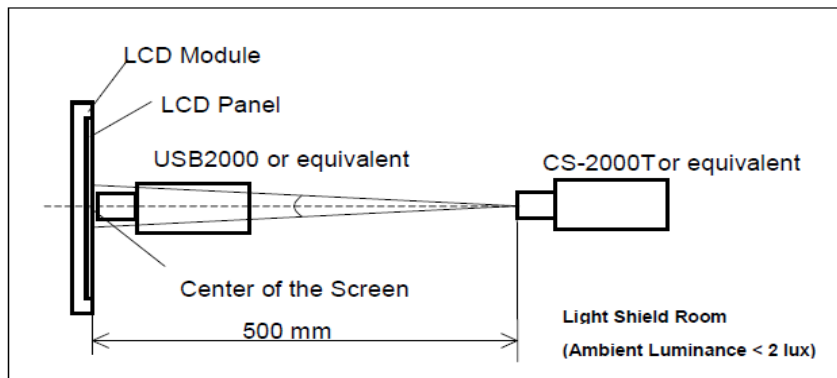
Measure the luminance of gray level 255 at center point

$L_{CP} = L(5)$

$L(x)$ is corresponding to the luminance of the point X at Figure in Note (6)

Note (5) Measurement Setup:

The LCD module should be stabilized at given temperature for 20 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 20 minutes in a windless room.



Note (6) Definition of White Variation (δW):

Measure the luminance of gray level 255 at 9 points

$$\delta W_{9p} = \{ \text{Minimum } [L(1) \sim L(9)] / \text{Maximum } [L(1) \sim L(9)] \} * 100\%$$



8.0 RELIABILITY

8.1 MTBF

The LCD module shall be designed to meet a minimum MTBF value of 50000 hours with normal, (25°C in the room without sunlight)

8.2 Tests

NO.	Test Item	Test condition	Criterion
1	High Temperature Storage	60°C±2°C 96H Restore 2H at 25°C Power off	
2	Low Temperature Storage	-10°C±2°C 96H Restore 2H at 25°C Power off	
3	High Temperature Operation	50°C±2°C 96H Restore 2H at 25°C Power on	
4	Low Temperature Operation	0°C±2°C 96H Restore 2H at 25°C Power on	
5	High Temperature & Humidity Operation	40°C±2°C 90%RH 96H Power on	
6	Temperature Cycle	-10°C ↔ 25°C ↔ 60°C 30min 5min 30min after 10cycle, Restore 2H at 25°C Power off	After testing, cosmetic and electrical defects should not happen.
7	Vibration Test	10Hz~45Hz, 100m/s ² , 120min	
8	Shock Test	Half-sine wave, 300m/s ² , 11ms	
9	Drop Test(package state)	800mm, concrete floor, 1 corner, 3 edges, 6 sides each time	1. After testing, cosmetic and electrical defects should not happen. 2. the product should remain at initial place 3. Product uncovered or package broken is not permitted.
10	Electro Static Discharge Test (non-operation)	150pF, 330Ω, Contact: ± 4KV, Air: ±8KV Measure point :LCD glass and metal bezel 200pF, 0Ω, ±200V contact test Measure point :IF connector pins	IEC61000-4-2: 2001 GB/T17626.2-2006



9.0 INSPECTION STANDARDS

9.1 Purpose

This incoming inspection standard shall be applied to TFT-LCD supplied by ZHONGSHEN to its customer.

9.2 Scope

This inspection standard contains Cosmetic Specifications and Electrical Specifications.

9.3 Classification of defects

9.3.1 Major defect.

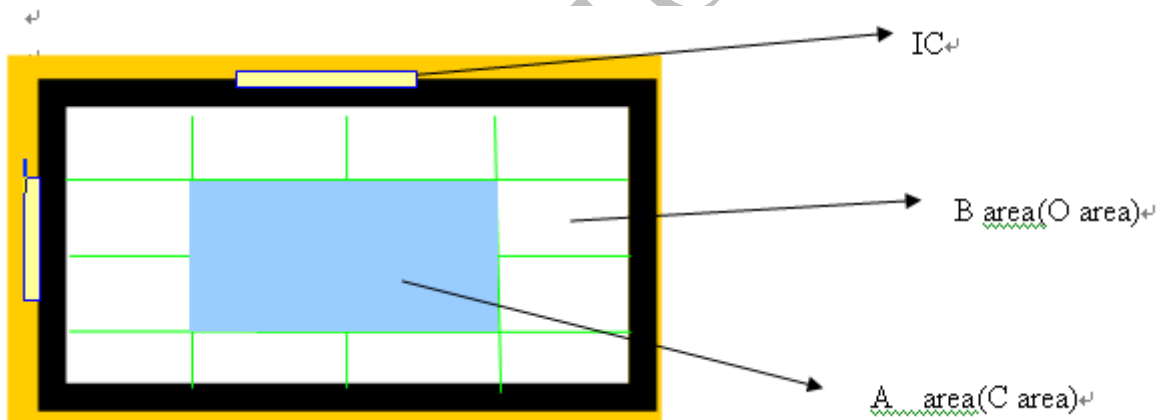
The major defect is a defect that is likely to result in product failure or reduction in Product's intended usage.

9.3.2 Minor defect.

The minor defect is a defect that has little bearing on the effective use or Operation of the product.

9.4 Definition

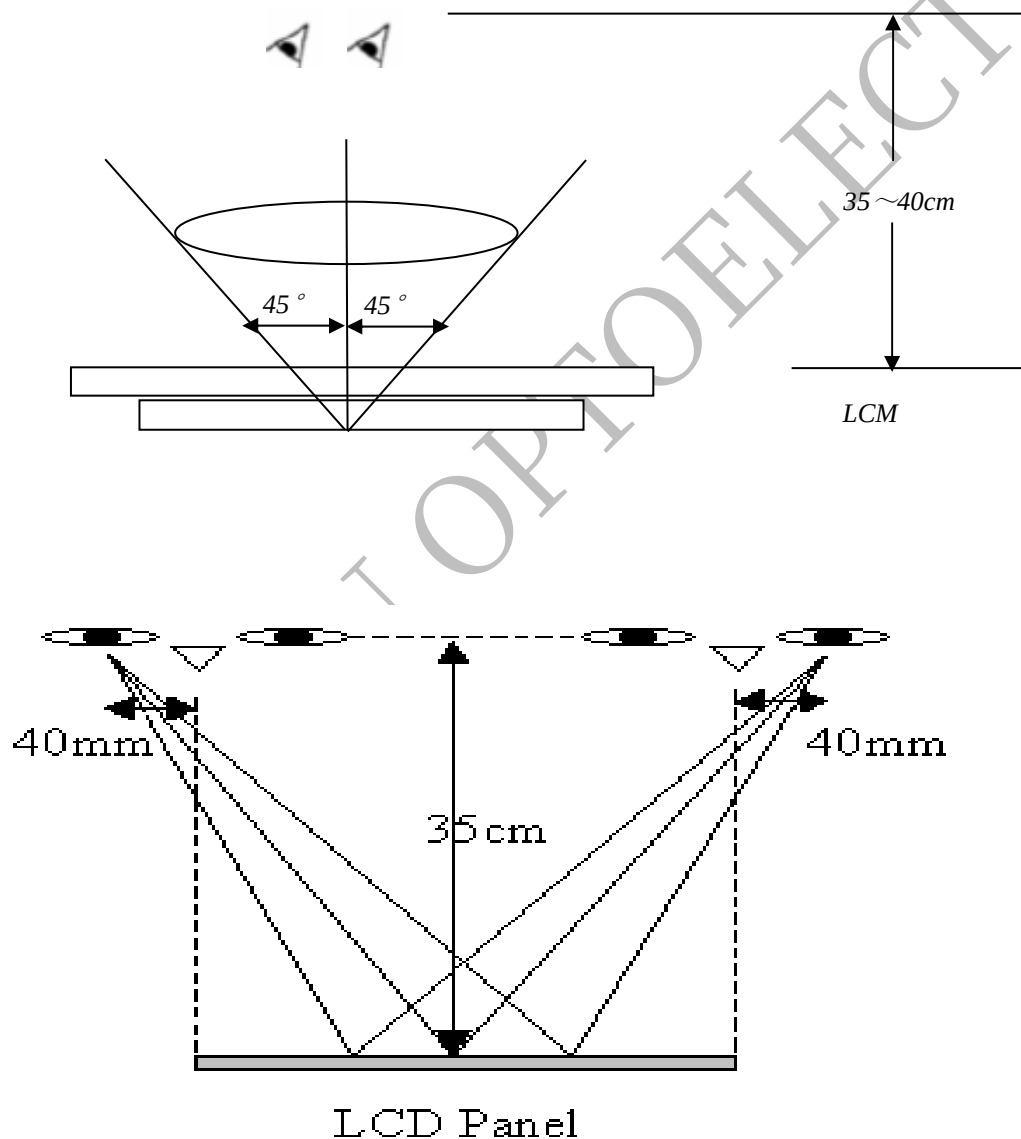
9.4.1 Display area definition





9.5 Inspection conditions is as follows

- 9.5.1 Viewing distance is approximately 35-40 cm
- 9.5.2 Viewing angle is normal to the LCD panel as 45°
- 9.5.3 Ambient temperature is approximately $25 \pm 5^\circ\text{C}$
- 9.5.4 Ambient humidity is $60 \pm 5\% \text{ RH}$
- 9.5.5 Ambient luminance is from 300-500 Lux.
- 9.5.6 Input signal timing should be typical value(3s-5s).
- 9.5.7 Mura & Light leakage inspection at ND-Filter 6%.





9.6 Sampling method

9.6.1 According to the MIL-STD-105E general inspection level , II Sampling plan.

9.6.2 AQL: MA 0.65 MI 1.0

9.7 Inspection Criteria

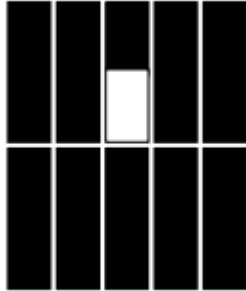
DEFECT TYPE			LIMIT			Defect	Note
VISUAL DEFECT	SCRATCH		W≤0. 05mm and L≤5mm		Ignore	Ma j	NOTE1
			0. 05mm<W≤0. 2mm L≤10mm		N≤4		
			10mm<L, 0. 1mm<W		N=0		
	INTERNAL	SPOT	Φ ≤0. 2mm		Ignore		
			0. 2mm< Φ ≤0. 5mm		N≤4		
			Φ >0. 5mm		N=0		
		FIBER	0. 1mm≤W≤0. 2mm L≤2. 5mm		N≤4		
			0. 2mm<W, 2. 5mm<L		N=0		
		POLARIZER BUBBLE	Φ ≤0. 25mm		Ignore		
			0. 25mm< Φ ≤0. 5mm		N≤4		
			Φ >0. 5mm		N=0		
		DENT	Φ <0. 25mm		Ignore		
			0. 25mm≤ Φ ≤0. 5mm		N≤4		
			Φ >0. 5mm		N=0		
ELECTRICAL DEFECT	BRIGHT DOT		C Area	O Area	Total	Ma j	NOTE2 NOTE3
			N≤4（contain C area and O area）		N≤4		
	DARK DOT		N≤5（contain C area and O area）		N≤5		
			TWO ADJACENT DOT		N≤1		
	THREE OR MORE ADJACENT DOT		NOT ALLOWLED				
	LINE DEFECT		NOT ALLOWLED				



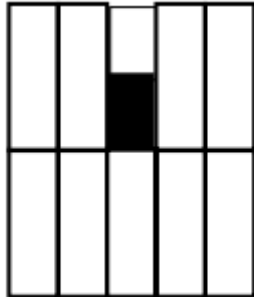
Note1: Minimum distance between dot defects and spot is 5mm;

Note2: The definition of Bright dot and Dark dot

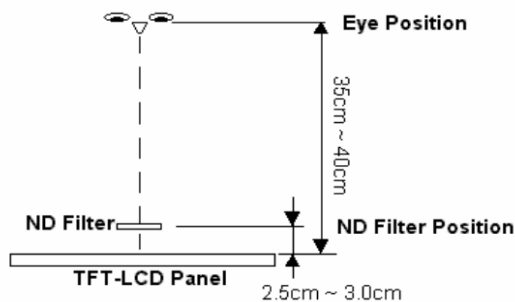
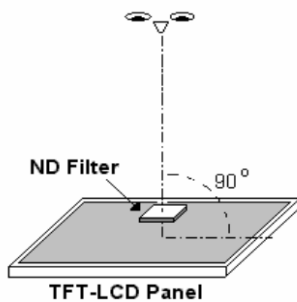
-bright area is more than 50% of one dot



-dark area is more than 50% of one dot



-The bright dot shall be visible under ND-Filter 5% as following:



NOTE3:

- A bit rate(bright dot model) $\leq 10\%$;
- Class Chipping but not affect the function of quality OK;
- Polarizing film appearance does not affect the function OK;

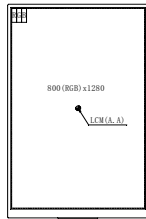


10.0 PACKING DRAWING

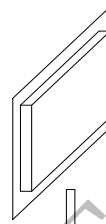


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Packing pictures:

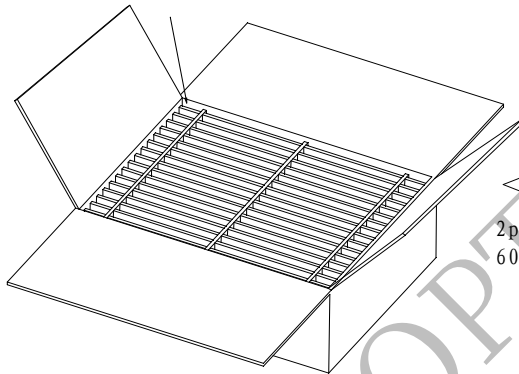


1PC/PE bagt

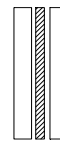


Remark: 2pcs LCM to be faced
By the middle of a foam
separated

In the box on 1pcs
foam board

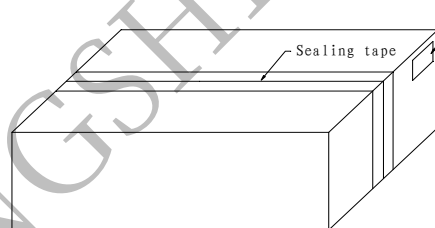


2pcsseperation/
60pcs/Carton



With foam separated

use tape to stick
sticker shipping mark



shipping mark

Sealing tape

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PAGE 页码	6/6				客户:			



11.0 HANDLING PRECAUTION

- (1) Don't disassemble and reassemble the module by self.
(禁止自行拆解)
- (2) Acid, alkali, alcohol or touched directly by hand will damage the display.
(酸性、碱性、酒精或手的直接接触将会损伤显示面)
- (3) Static electricity will damage the module. Please configure grounding device.
(静电会损伤模组，请装配接地设备)
- (4) The strong vibration, shock, twist or bend will cause material damage, even module broken.
(强烈的撞击、震动、扭转或弯曲将会造成原材损伤，甚至面板破裂)
- (5) It is easy to cause image sticking while displaying the same pattern for very long time.
(长期显示同一画面会造成影像残留)
- (6) The response time, brightness and performance will vary from different temperature.
(响应时间、亮度与均匀性会因温度而有所改变)
- (7) Starting from the date of shipment in the photoelectric products for a period of 12 months.
(从中深光电出货之日开始产品保质期为 **12** 个月)